

```

path: /etc/httpd/conf/httpd.conf
regexp: '^Listen\s+80$'
line: 'Listen 0.0.0.0:80'

- name: create_vhost
  copy:
    content: |
      <VirtualHost *:80>
        ServerName helloworld.myexample.com
        DocumentRoot /var/www/html
        ErrorLog /var/log/httpd/error.log
        CustomLog /var/log/httpd/access.log combined
        <Directory /var/www/html>
          AllowOverride All
          Require all granted
        </Directory>
      </VirtualHost>
    dest: /etc/httpd/conf.d/helloworld.conf

- name: create_webroot
  file:
    path: /var/www/html
    state: directory
    owner: apache
    group: apache
    mode: '0755'

- name: create_index_html
  copy:
    content: '<h1>Hello, World!</h1>'
    dest: /var/www/html/index.html
    owner: apache
    group: apache
    mode: '0644'

- name: start_httpd
  service:
    name: httpd
    state: started
    enabled: true

- name: install_testinfra
  ansible.builtin.pip:
    name: pytest-testinfra
    executable: pip3

```

Define the Testinfra test cases: Once the templating part is done, we will need to write a simple test case using Testinfra to ensure the AMI has 'httpd' package successfully installed, configured, and running fine. In short, the AMI should be in the desired state. Here is an example of a sample test case for 'httpd' service:

```

import testinfra

def test_apache_is_installed(host):
    assert host.package("httpd").is_installed

def test_apache_is_running(host):
    assert host.service("httpd").is_running
    assert host.service("httpd").is_enabled

def test_apache_virtual_host(host):
    virtual_host = """
    <VirtualHost *:80>
        ServerName helloworld.myexample.com
        DocumentRoot /var/www/helloworld
        ErrorLog /var/log/httpd/helloworld-error.log
        CustomLog /var/log/httpd/helloworld-access.log
    combined
        <Directory /var/www/helloworld>
            AllowOverride All
            Require all granted
        </Directory>
    </VirtualHost>
    """
    assert host.file("/etc/httpd/conf.d/helloworld.conf").exists
    assert host.file("/etc/httpd/conf.d/helloworld.conf").contains(virtual_host)

def test_apache_index_html(host):
    assert host.file("/var/www/html/index.html").exists
    assert host.file("/var/www/html/index.html").contains("<h1>Hello, World!</h1>")

```

Bake the golden AMI: Now, we can start baking the AMI once we are done with all the steps mentioned above. So let's run the following command, which will initiate the baking process:

```
$ packer build -var-file=templates/webserver.pkrvars.hcl templates/webserver.pkr.hcl
```

amazon-ebs.webserver: output will be in this color.

```
==> amazon-ebs.webserver: Prevalidating any provided VPC information
```

```
==> amazon-ebs.webserver: Prevalidating AMI Name: webserver-2.4-20230520103749
```

```
amazon-ebs.webserver: Found Image ID: ami-028a473e64001bfb2
```

```
==> amazon-ebs.webserver: Provisioning with Ansible...
```

```
amazon-ebs.webserver: Setting up proxy adapter for Ansible...
```

```
*****
```